

Quadrature All-Pass Topologies for UHJ

A Multi-Rate Comparative Study

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Abstract

This study compares the all-pass network topologies that realise a wideband 90-degree quadrature for UHJ encoding. It measures phase accuracy, computational cost, sample-rate behaviour, and group delay on one harness, and it recommends a topology for a standalone quadrature plugin.

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1 Introduction

The UHJ encoding matrix requires a wideband 90-degree phase shift — the j operator — applied across the full audible spectrum with sub-degree accuracy. Two questions organise this study: which all-pass topology delivers the greatest phase accuracy per unit computational cost, and how each topology behaves as the sample rate changes from 44.1 kHz to 192 kHz.

The problem has a continuous history. Analog phase-difference networks, analysed by Bedrosian [1], showed that cascaded first- and second-order all-pass sections can approximate a 90-degree differential phase over a prescribed bandwidth. Schroeder’s 1958 differential all-pass arrangement is the conceptual origin of transforming a mono signal into a perceived stereo image through phase manipulation; its design goal was decorrelation rather than a held quadrature relationship, yet it established the building block that later work would repurpose. Gerzon applied the quadrature idea to UHJ surround encoding [2], where a precise 90-degree shift converts mid-side signals into the spatially encoded B-format lateral component. The modern digital realisation rests on the lattice all-pass section studied by Regalia, Mitra, and Vaidyanathan [3], which provides a numerically well-conditioned structure with a single tunable coefficient.

This study contributes a single evaluation harness that places every topology on one footing across four metrics and produces an evidence-based recommendation for the quadrature processor inside a standalone UHJ plugin.

2 Topologies Under Test

2.1 RBJ Second-Order Biquad Cascade

The RBJ all-pass biquad [4] realises the continuous-time prototype

$$H(s) = \frac{s^2 - (\omega_0/Q)s + \omega_0^2}{s^2 + (\omega_0/Q)s + \omega_0^2}, \quad (1)$$

bilinearly transformed to the z -domain. A cascade of such sections, with centre frequencies and Q values chosen by a minimax phase-error optimisation at the target sample rate, approximates the 90-degree differential phase across the desired band. This is the topology currently employed in the `x2uhj` plugin, where coefficients are redesigned whenever the host sample rate changes.

2.2 First-Order Polyphase (Niemitalo)

The first-order polyphase all-pass section [5, 6] has the transfer function

$$H(z) = \frac{a^2 - z^{-2}}{1 - a^2 z^{-2}}, \quad (2)$$

requiring one multiply per section after the coefficient a is fixed. Two parallel paths, each a cascade of such sections with complementary coefficients, deliver the 90-degree quadrature pair. The WigWare distribution [7] ships a widely cited set of fixed coefficients designed for 44.1 kHz; this study evaluates those published coefficients as one realisation mode and a per-rate minimax redesign as the second.

2.3 Elliptic Half-Band Derived Hilbert (Harris–Berdahl–Abel)

Harris, Berdahl, and Abel [8] derive a Hilbert transformer by splitting an elliptic half-band lowpass filter into two parallel all-pass branches whose outputs are in quadrature. The prototype is designed in normalised frequency against an elliptic approximation [9], giving steep transition-band attenuation with a low filter order. The two branch outputs form the in-phase and quadrature signals directly, making this a structurally efficient realisation for hardware or fixed-point contexts.

2.4 Historical Context: Schroeder Differential All-Pass

Schroeder’s 1958 differential all-pass network is the conceptual ancestor of all the topologies above. Its purpose was decorrelation: by splitting a mono signal into two paths that share identical magnitude responses but differ in phase, it creates the perception of spaciousness without altering spectral balance. The study treats it as historical framing rather than as a competitor, because its design target — maximum decorrelation over frequency — differs from the present goal of a held 90-degree phase difference across the audio band.

3 Method

3.1 Uniform Evaluation Interface

Every topology is accessed through a common interface that exposes three operations: **design**, which accepts a sample rate and returns a coefficient set; **phase**, which evaluates the differential phase response at a specified frequency grid; and **cost**, which returns the multiply count per output sample. The uniform interface ensures that the harness logic is topology-agnostic and that comparisons reflect only the properties of the all-pass networks themselves.

3.2 Realisation Modes

Each topology is exercised in two realisation modes. In the *fixed-coefficient* mode, coefficients are designed once at a single reference rate (44.1 kHz for the Niemitalo polyphase, using the published WigWare values [7]) and applied unchanged at every test rate. In the *per-rate* mode, the minimax optimisation runs afresh at each sample rate, yielding the best achievable accuracy for that rate. Comparing the two modes reveals how much accuracy degrades when a fixed design is used outside its intended rate.

3.3 Metrics

Four metrics are recorded for each topology and realisation mode. *Maximum phase error* quantifies the worst-case deviation from 90 degrees across the evaluation band, reported in degrees; the companion *error versus order* curve shows how accuracy scales with the number of all-pass sections. *Sample-rate robustness* measures the maximum phase error at each of the six standard rates: 44.1, 48, 88.2, 96, 176.4, and 192 kHz. *Computational cost* counts the real multiplies required to produce one pair of quadrature output samples, enabling a cost-normalised accuracy comparison. *Differential group delay* records the deviation of the group-delay difference between the two quadrature paths from a flat 0 s response, which affects the perceived image stability of encoded material.

3.4 Minimax All-Pass Design

Coefficient optimisation follows the established minimax all-pass design method described by Lang [10], which minimises the Chebyshev norm of the phase error over the target band subject to the all-pass magnitude constraint.

References

- [1] S. D. Bedrosian, “Normalized design of 90-degree phase-difference networks,” *IRE Transactions on Circuit Theory*, vol. 7, no. 2, pp. 128–136, 1960.
- [2] M. A. Gerzon, “Ambisonics in multichannel broadcasting and video,” *Journal of the Audio Engineering Society*, vol. 33, no. 11, pp. 859–871, 1985, aES Preprint 2034, 74th Convention, 1983.
- [3] P. A. Regalia, S. K. Mitra, and P. P. Vaidyanathan, “The digital all-pass filter: A versatile signal processing building block,” *Proceedings of the IEEE*, vol. 76, no. 1, pp. 19–37, 1988.
- [4] R. Bristow-Johnson, “Cookbook formulae for audio EQ biquad filter coefficients,” Audio EQ Cookbook.
- [5] O. Niemitalo, “90 degree phase difference IIR allpass pair,” <https://yehar.com/blog/?p=368>, 2003.
- [6] R. Ansari, “IIR discrete-time hilbert transformers,” *IEEE Transactions on Acoustics, Speech, and Signal Processing*, vol. 35, no. 8, pp. 1116–1119, 1987.
- [7] B. Wiggins, “WigWare UHJ encoder/decoder (Reaper JSFX),” <https://www.brucewiggins.co.uk/>.
- [8] D. Harris, E. Berdahl, and J. S. Abel, “An infinite impulse response (IIR) hilbert transformer filter design technique for audio,” in *Proc. 129th Audio Engineering Society Convention*, 2010, paper 8258.

- [9] H. W. Schüßler and P. Steffen, “Halfband filters and hilbert transformers,” *Circuits, Systems and Signal Processing*, vol. 17, no. 2, pp. 137–164, 1998.
- [10] M. Lang, “Allpass filter design and applications,” *IEEE Transactions on Signal Processing*, vol. 46, no. 9, pp. 2505–2514, 1998.